CHITRANG POTDAR

Backend Engineer

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PROFESSIONAL SUMMARY

Backend Engineer working with Python across backend systems, automation, and AI-integrated workflows. Experience includes building APIs, data pipelines, and LLM-based systems, with a focus on designing systems that are maintainable, reliable, and scale effectively. Built and maintained production systems across multiple domains with measurable impact.

TECHNICAL SKILLS

Languages & Core: Python, SQL, System Design
Frameworks & APIs: FastAPI, Flask, REST APIs
Databases & Caching: PostgreSQL, Redis, MSSQL
AI / LLM / Systems: Claude API, LlamaIndex, CrewAI, AWS Transcribe, Google Cloud STT, Whisper, Scikit-learn, MLflow, ZenML
Automation & Testing: Playwright, Pytest
Cloud & Infrastructure: AWS (Lambda, API Gateway, S3, EC2), Docker, GitHub Actions, CI/CD

PROFESSIONAL EXPERIENCE

Backend Engineer — Oct 2025 – Present

ArrayPointer Pvt. Ltd., Pune

Investment Operations Platform

Flask · PostgreSQL · Redis · Docker · GitHub Actions · AWS EC2 · Pytest · Playwright

- Built a multi-tenant backend with strict query-level data isolation across users, portfolios, and organizations in a financial platform powering internal operational systems.
- Developed REST APIs for portfolio review, rejected order tracking, and financial calculators; integrated Jira API for automated issue tracking.
- Reduced redundant DB queries via Redis-based session caching; maintained 80%+ unit test coverage with Pytest.
- Automated NSE rejected order processing and E2E client workflows using Playwright, saving ~2–3 hours/week; set up CI/CD via GitHub Actions and Docker.

AI Meeting Intelligence Pipeline

AWS Lambda · API Gateway · S3 · AWS Transcribe · Claude API

- Designed async serverless pipeline around API Gateway's 29s timeout — audio + agenda from S3 → AWS Transcribe → Claude API → structured output (summary, agenda scoring, action items).

AI Voice Conversational System with RAG

FastAPI · Google Cloud STT / Whisper · TTS · Claude API · LlamaIndex

- Built voice AI assistant pipeline: STT → RAG retrieval → LLM → TTS with semantic chunking and embedding-based retrieval for context-grounded responses.
- Designed modular architecture allowing knowledge base updates without modifying core pipeline logic.

Python Developer Intern — Dec 2023 – Apr 2024

ArrayPointer Pvt. Ltd., Pune

- Automated PDF parsing (electricity bills, CAS files) using PyPDF2, Camelot, Tabula — reduced manual extraction effort by ~70% with structured JSON output.
- Built XGBoost revenue forecasting model with incremental learning — improved forecast accuracy by ~20% as new data was ingested.

PROJECTS

TaskHub – Multi-Tenant Project Management Backend Aug – Sep 2025

github.com/Dev-Chitrang/TaskHub-Full-Stack-Project

FastAPI · PostgreSQL · Docker · GitHub Actions · Pytest

- Built a SaaS-style project management backend from scratch — workspace isolation, RBAC, and full auth flow (email verification, 2FA via TOTP, password recovery).
- Containerised with Docker, deployed via GitHub Actions CI/CD, with Pytest coverage across all core API modules.

Retail DataOps & MLOps Platform Mar – Jul 2025

github.com/Dev-Chitrang/data-warehouse-to-mlops

FastAPI · MSSQL · Scikit-learn · ZenML · MLflow · NeuralProphet

- Designed ETL pipelines using medallion architecture (Bronze/Silver/Gold) on SQL Server for structured data processing at scale.
- Exposed demand forecasting, customer segmentation, and recommendation models via FastAPI; managed full ML lifecycle with MLflow and ZenML.

AI Trip Planner – Multi-Agent Orchestration System Sep – Oct 2025

github.com/Dev-Chitrang/Multi-Agent-Trip-Planner

CrewAI · Gemini 2.0 Flash · SerpAPI · Python

- Designed a multi-agent system with three specialised agents (Flight, Hotel, Itinerary) — each with a defined role, toolset, and reasoning layer powered by Gemini 2.0 Flash.
- Integrated SerpAPI for real-time flight and hotel data, enabling the system to plan trips using live availability.

EDUCATION

B.E. Computer Engineering Jun 2020 – Jun 2025

Savitribai Phule Pune University, Pune

CERTIFICATIONS

- IBM AI & Cloud — AICTE Certified
- VOIS AI & Data Analysis — AICTE Certified

RESEARCH & PUBLICATIONS

Corporate Credit Rating Analysis using AI & Sentiment Analysis Nov 2024

IJIRCCE | Impact Factor: 8.65

github.com/Dev-Chitrang/Final-year-project/tree/Backend